

Cooling System Design for FSAE Electric Vehicle

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Abstract

This internal report establishes an evidence-led thermal-management baseline for a rear-drive electric Formula Student powertrain comprising one AMK KW26-S5-FSE-4Q inverter and two DYNASYN DD5-14-10 motors. FSG 2026 and AMK primary documentation separate verified component limits from engineering assumptions. The numerical method couples DD5 loss-map sensitivity, KW26 liquid-heat assumptions, two digitized Boyd 6310G3 heat exchangers, exact SPAL VA99 fan data and a labelled CWA150 pump curve. A one-dimensional finite-volume coolant solver then integrates advection, discrete heat sources, transient storage and distributed radiator rejection. Its 20–160-cell and time-step studies pass a 0.1 K convergence gate and close the semidiscrete energy balance to 3.2×10^{-12} W. Under 40°C E3 air, the middle 3.062 kW case converges to 59.45°C KW26 inlet; the legacy 6.0 kW/10 s boundary raises the peak to 61.08°C. Theory, steady curve coupling and transient numerical heat transfer therefore agree: the passive Boyd/SPAL/CWA combination is a procurement no-go for E3. Solid-component conjugate heat transfer remains outside the claim because internal geometry and material/contact data are unavailable.

Keywords: Formula SAE, Electric Vehicle, Cooling System, Radiator, Fan, Pump, Catch Can, Swirl Pot, Hoses, FSAE Rules

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1 Candidate Cooling-System Topology

The current internal baseline is a closed liquid loop for one inverter and two motors. The topology below is a **candidate architecture**, not an installation drawing or an approved bill of materials. It is sufficient to establish the governing flow constraints: the AMK inverter requires at least 10 L/min and each DD5 branch requires at least 4 L/min at its rated cooling condition.

The candidate uses an inverter in series with parallel motor branches and parallel radiator branches. Equal branch flow is an initial arithmetic assumption only; the final splitter geometry and actual flow split require system-curve calculation and bench measurement. The dual-sidepod layout is similarly a packaging hypothesis pending local sidepod/duct CAD and air-path evidence.

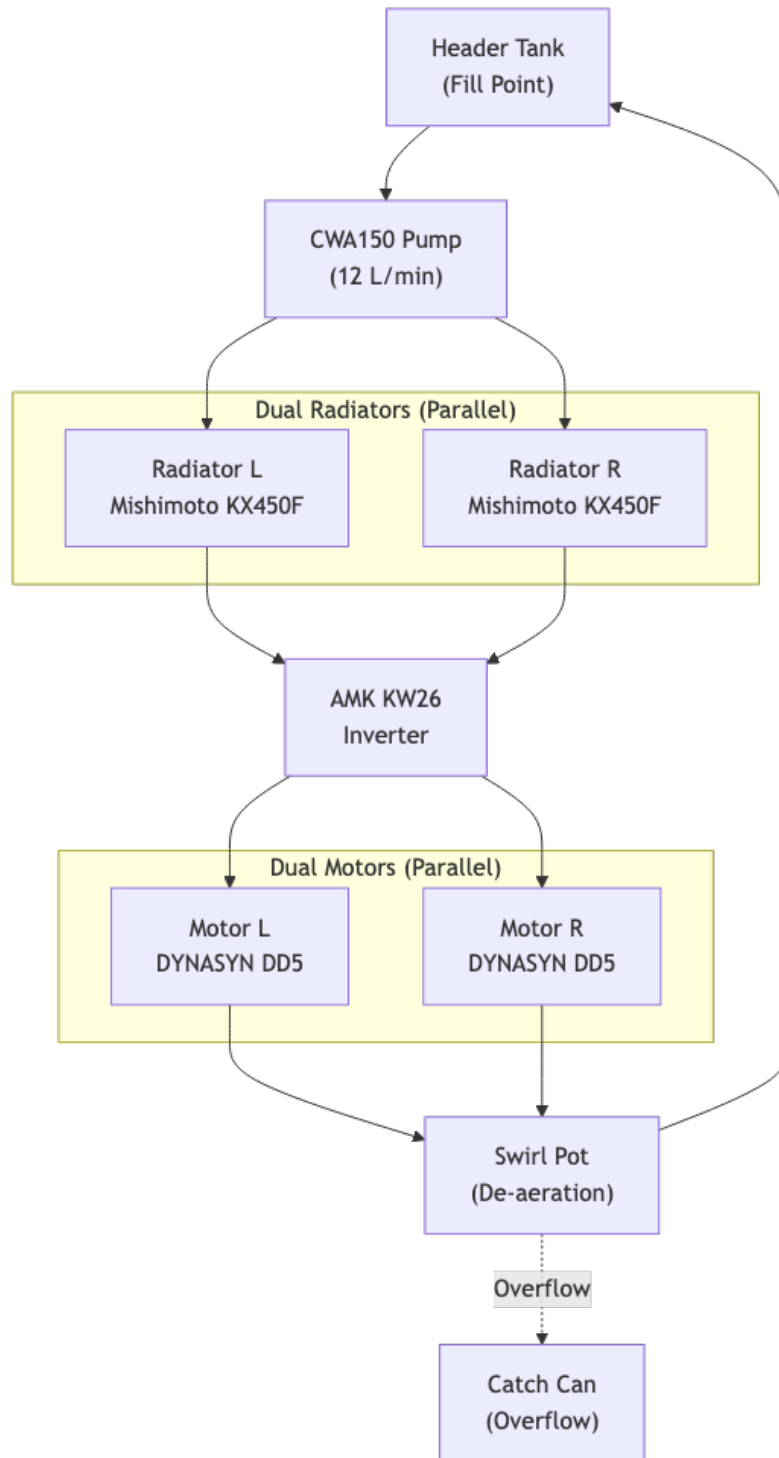


Figure 1: Candidate logical flowchart of the VED cooling circuit.

The figure does not select a pump, radiator, header tank or swirl pot. The final fill/degassing hardware will be chosen only after the hydraulic, fill/bleed and overflow tests defined in Sec-

tion 11.

2 Competition Compliance Baseline

Status: not a final compliance approval. The legacy rule table mixed revisions and contained unsupported claims. This revision uses Formula Student Germany (FSG) Rules 2026 v1.1 as an initial baseline only. The entered event and its final rule revision must be recorded before any procurement or technical inspection claim is made. The authoritative source is the FSG rulebook¹.

Table 1: Compliance items to verify against the entered-event rules

FSG reference	Requirement to carry into design	Evidence required before release
T7.2.2–T7.2.9	Cooling medium, driver shielding, sealing, catch-can capacity/-mounting, temperature rating, and discharge routing are regulated. FSG 2026 permits plain water, air, or oil; it does not support the legacy “pure water only” statement.	Event rule revision; schematic; system volume calculation; component temperature ratings; inspection photographs.
EV5.5.16	Tractive-system accumulator cooling/ventilation openings have constraints; cooling architecture must preserve the accumulator enclosure requirements.	Accumulator wall/opening drawing and event-specific compliance review.
EV5.8.3–EV5.8.4	Thermally critical cells require monitoring. The applicable cell limit is 60°C or the cell datasheet limit, whichever is lower.	Cell datasheet, sensor locations, BMS logic, and validation log.
IN9.2	Rain-test requirements affect fan/pump operation and electrical integration.	Event inspection checklist, wiring diagram, and witnessed test result.

No generic fan-power, IP-rating, or noise limit is asserted here: each must be demonstrated by the applicable event handbook and the selected component documentation. The single-loop/multi-loop choice is an engineering trade-off, not a rule conclusion; it remains pending thermal and hydraulic verification.

¹Formula Student Germany, [Rules 2026 v1.1](#), checked 9 August 2026.

3 Powertrain Cooling Requirements and Data Status

The vehicle architecture is one AMK KW26-S5-FSE-4Q inverter with two DYNASYN DD5-14-10 motors. The circuit is a candidate single-loop architecture; its final topology, component selections and losses are not approved in this revision.

3.1 Verified manufacturer limits

Table 2: Primary-source cooling constraints

Item	Verified limit and interpretation
AMK inverter	Rated 26 kVA. Its liquid-cooling rated condition is coolant flow temperature $\leq 25^\circ\text{C}$, flow ≥ 10 L/min, pressure 1.5 bar and coolant rise < 5 K. The cold plate can dissipate up to 2,000 W at that condition; this is not an inverter-loss map.
DYNASYN DD5 motor (each)	Rated power 12.3 kW, rated torque 9.8 Nm at 12,000 rpm, maximum torque 21 Nm. The user must design the jacket. Rated cooling requires coolant flow temperature $\leq 40^\circ\text{C}$, flow ≥ 4 L/min and coolant rise < 5 K; derating is 1%/K from 40 to 60°C .
Circuit implication	The inverter's 10 L/min requirement governs total loop flow. With two equal motor branches this nominally gives 5 L/min per motor, but the actual split must be demonstrated by a hydraulic calculation and bench measurement.

Sources are the AMK RACING KIT manual PDK_205481 (sections 6.2.1–6.2.2) and AMK DD5 data sheet A2370DD, retained in the internal evidence dossier.

3.2 Heat-load model: unresolved inputs, not fixed facts

The legacy report used $26 \text{ kVA} \times 2\%$ for inverter loss and 26 kW per motor at 95% efficiency. Neither is an acceptable loss derivation: kVA is not a real-power operating point and the supplied DD5 rated power is 12.3 kW. AMK's archived DD5 workbooks do contain 80/100/120°C total-loss maps by speed and torque; the model now reads those maps directly. For example, the 80°C map interpolates to approximately 1.37 kW total motor loss at 12,000 rpm and 9.8 Nm. The map temperature definition and the fraction of total motor loss transferred to the liquid jacket remain to be established. The previous 3.256 kW continuous and 6.0 kW peak values are therefore retained only as **legacy screening hypotheses**.

For each recorded drive-cycle point, the revised model shall calculate

$$\dot{Q}_{total} = \dot{Q}_{inverter}(V, I, f_{sw}, T_{c,in}, \dot{m}) + 2 \dot{Q}_{motor}(n, \tau, T_{c,in}, \dot{m}), \quad (1)$$

using manufacturer loss maps or measured calorimetry. The DD5 map supplies total motor loss; thermal partitioning to coolant requires a jacket thermal model or test. The inverter loss map remains required. Until these inputs exist, no component is sized from a claimed thermal load.

4 Evidence-Limited Thermal and Hydraulic Baseline

4.1 Purpose and boundary

This section replaces the previous apparent precision in radiator UA , fin area, air velocity and heat-load calculations with a reproducible screening model. The model is stored in `sim/zero_d/` with its inputs and tests. Verified AMK limits remain separate from digitized manufacturer curves and explicitly labelled pump, duct, loss-partition and curve-reading uncertainties.

4.2 Minimum liquid-side capacity

At the inverter-governed minimum flow rate of 10 L/min, using nominal water properties $\rho = 992 \text{ kg/m}^3$ and $c_p = 4180 \text{ J/(kg K)}$, the coolant heat-capacity rate is

$$C_c = \dot{m}c_p = 691 \text{ W/K.} \quad (2)$$

The liquid-side screening heat transport associated with a 5 K rise is therefore

$$\dot{Q}_{5K} = C_c \Delta T = 3.46 \text{ kW.} \quad (3)$$

This is neither radiator capacity nor powertrain heat generation. It is only a necessary water-side check at the stated flow.

Table 3: Legacy-load screening at 10 L/min

Scenario	Legacy heat hypothesis	Coolant rise	5 K screen
Continuous	3.256 kW	4.71 K	passes
Peak, 10 s	6.000 kW	8.68 K	fails

The table must not be read as validation of either heat hypothesis. It demonstrates why transient thermal capacitance, component temperatures and flow distribution must be measured before claiming that a peak event is safe.

4.3 Strict high-ambient design condition

The initial strict screen uses 40°C ambient and no refrigeration. A passive radiator cannot deliver coolant below the ambient temperature in steady operation. It follows directly that the AMK

inverter's 25°C rated coolant-inlet condition is **not feasible** in this high-ambient passive case. The prior conclusion that a passive system supports rated operation at 40–45°C is withdrawn.

The internal design study must consequently report distinct operating regimes:

1. rated manufacturer cooling condition (25°C inverter inlet),
2. high-ambient passive operation with the applicable component derating explicitly quantified, and
3. transient peak operation, bounded by measured thermal capacitance and temperature limits.

The internal operating envelope also includes zero-speed heat soak and cold/condensation protection. The 40°C high-ambient scenarios are conservative engineering cases rather than FSG requirements. Each future model or test result shall identify the applicable envelope, source of every boundary condition and hardware revision.

4.4 Governing theory and falsifiable predictions

For a steady liquid loop, conservation of energy gives

$$C_c = \dot{m}c_p, \quad T_{\text{rad,in}} - T_{\text{rad,out}} = \frac{\dot{Q}_{\text{liq}}}{C_c}. \quad (4)$$

The Boyd performance ordinate is a heat-exchanger conductance based on the initial temperature difference, $K_{\text{HX}} = \dot{Q}/(T_{\text{rad,in}} - T_{\text{air,in}})$, rather than an assumed vehicle UA . Consequently,

$$T_{\text{rad,in}} = T_{\text{air,in}} + \frac{\dot{Q}_{\text{liq}}}{K_{\text{HX}}}. \quad (5)$$

For the candidate series order radiator–inverter–parallel motors, the component inlet temperatures are

$$T_{\text{inv,in}} = T_{\text{rad,out}}, \quad T_{\text{mot,in}} = T_{\text{inv,in}} + \frac{\dot{Q}_{\text{inv,liq}}}{C_c}. \quad (6)$$

These equations predict, before any component curve is fitted, that a passive steady loop with positive heat input cannot supply sub-ambient coolant. At 40°C radiator-inlet air it therefore cannot reproduce the KW26 rated 25°C supply condition. Because the KW26 manual also requires the cold plate surface to remain below 40°C, any numerical E3 result with $T_{\text{inv,in}} \geq 40^\circ\text{C}$ disproves that candidate duty point without needing a detailed cold-plate model.

The air-side operating point must satisfy

$$\Delta p_{\text{fan}}(\dot{V}_a, n) = \Delta p_{\text{core+duct}}(\dot{V}_a). \quad (7)$$

Where only the maximum-speed fan curve is published, the fan affinity laws $\dot{V}_a \propto n$ and $\Delta p \propto n^2$ are used only as a labelled sensitivity approximation inside the transformed published curve range. Likewise, the liquid operating point is the intersection of the pump curve and the circuit

curve. Quadratic branch-loss sensitivity is written as $\Delta p_i = K_i \dot{V}_i^2$, with equal pressure drop and conservation of flow across parallel branches. A numerical candidate must therefore close energy, air flow, total liquid flow and motor-branch flow simultaneously; passing a stand-alone catalogue-capacity calculation is insufficient.

4.5 Coupled numerical result, not physical validation

The archived DD5 workbook gives a total loss of 1.374 kW per motor at 12,000 rpm, 9.8 Nm and its 80°C map label. The numerical study sweeps DD5 liquid fractions of 0.50, 0.75 and 1.00 and KW26 cold-plate loads of 0.5, 1.0 and 2.0 kW. The selected numerical screen uses two parallel Boyd 6310G3 cores, one SPAL VA99 fan per core at a 0.35 speed ratio, an air-system pressure factor of 1.5, and the CWA150 sensitivity curve. These are calculation boundaries, not installed measurements.

The nominal fan/core intersection is 6.56 m³/min per radiator at 45.6 Pa. Subtracting the recorded 10 Pa fan-curve reading uncertainty moves the intersection below the transformed published fan-curve range, so the air point is not robustly bounded. On the liquid side, subtracting 0.1 bar from the pump curve, adding 0.03 bar to each radiator curve and assuming 1.5 bar non-radiator loss at 10 L/min gives 10.03 L/min total flow. A 2:1 motor-branch resistance ratio leaves 4.16 L/min in the lower-flow branch. The nominal hydraulic requirements are therefore met by only 0.03 L/min total-flow and 0.16 L/min branch-flow margins; this is not a robust approval.

Table 4: Selected E3 coupled thermal screen

DD5 liquid	KW26 heat	Total heat	Coolant rise	KW26 inlet	Result
50%	0.5 kW	1.874 kW	2.70 K	51.9°C	fail
75%	1.0 kW	3.062 kW	4.41 K	59.4°C	fail
100%	2.0 kW	4.749 kW	6.85 K	70.1°C	fail

Every heat-partition combination fails the E3 conditional-procurement gate. Even the lowest load predicts 54.6°C radiator hot inlet, 51.9°C KW26 coolant inlet and 52.6°C DD5 inlet. The KW26 inlet is already above the manual's 40°C maximum cold-plate surface temperature, so no detailed cold-plate model can reverse the rejection. The legacy 3.256 kW continuous boundary predicts 65.3°C radiator hot inlet and 60.6°C KW26 inlet. The legacy 6.0 kW/10 s value is retained only as a coolant-domain transient boundary; it is not a component-temperature prediction because solid thermal masses and contact resistances are absent.

The coupled calculation verifies the theoretical high-ambient no-go; it does not physically validate the vehicle. The current Boyd/SPAL/CWA combination shall not be procured as the final E3 passive cooling system. A new candidate may enter the procurement gate only after changing

the infeasible boundary (for example a lower declared ambient/duty or active refrigeration) and re-running the same coupled checks. Post-procurement testing is calibration, not permission to rescue a candidate that already fails numerically.

4.6 Finite-volume transient heat-transfer verification

The steady map calculation is independently reproduced with a one-dimensional finite-volume coolant energy equation,

$$\rho V_i c_p \frac{dT_i}{dt} = C_c(T_{i-1} - T_i) + \dot{Q}_i - (UA)_i(T_i - T_{\text{air}}). \quad (8)$$

Upwind advection transports heat around the closed loop; the KW26 and combined DD5 liquid heat are applied at separate cells, and radiator rejection is distributed over the exchanger cells. To preserve the Boyd $K_{\text{HX}} = \dot{Q}/(T_{\text{rad,in}} - T_{\text{air,in}})$ map, the equivalent distributed conductance is

$$UA_{\text{eq}} = -C_c \ln \left(1 - \frac{K_{\text{HX}}}{C_c} \right). \quad (9)$$

The selected run uses 80 cells, 2.0 L assumed loop coolant volume, 10.03 L/min flow, a CFL number of 0.25 and a 0.0374 s nominal time step. It first integrates the 3.062 kW middle sensitivity case for 600 s, then applies the legacy 6.0 kW boundary for 10 s with 2.0 kW assigned to the KW26 liquid path. The baseline converges to 59.45°C KW26 inlet, 60.90°C DD5 inlet and 63.87°C radiator inlet. During the peak, the respective maxima are 61.08°C, 62.43°C and 68.11°C. Sweeping 1.5–3.0 L loop volume changes the peak KW26 inlet from 62.78 to 59.44°C; thermal storage delays the excursion but cannot restore the E3 limit.

Table 5: Finite-volume mesh convergence for the 10 s peak

Cells	Time step	Peak KW26 inlet
20	0.1495 s	61.439°C
40	0.0747 s	61.183°C
80	0.0374 s	61.078°C
160	0.0187 s	61.108°C

The 80–160-cell difference is 0.0294 K and the two finest time-step results differ by 0.0183 K, passing the internal 0.1 K numerical gate. The maximum semidiscrete energy-balance residual is 3.2×10^{-12} W. The finite-volume steady radiator inlet differs from the map equation by 0.078 K. The transient solver therefore verifies energy conservation, storage, advection, radiator rejection and numerical convergence. It is not solid-component conjugate heat transfer; OpenFOAM CHT still requires internal geometry, materials and contact data. The converged coolant-domain result strengthens, rather than replaces, the procurement no-go.

4.7 Remaining evidence limits

The Boyd curves and exact SPAL maximum-speed curve are manufacturer-issued. The reduced-speed fan curve uses affinity-law scaling, and the complete CWA150 curve remains a secondary-hosted sensitivity input. Exact hose, fitting, KW26 and DD5 jacket pressure losses are represented by the declared system-curve range. A final positive procurement decision still requires a candidate that passes without extrapolation, a quantified KW26 high-inlet operating limit, and a real drive cycle for E5.

5 Radiator Selection: Numerical No-Go

The former Mishimoto, Setrab and generic motorcycle-radiator conclusions remain withdrawn because no exact heat-rejection and two-sided pressure-drop maps were available. The current study instead evaluated two exact manufacturer-mapped Boyd candidates.

Table 6: Radiator candidate status

Candidate	Model-usable evidence	Decision
Boyd 6210G1	Manufacturer Q/initial-temperature-difference map and exact air/liquid pressure-drop curves; 2.3 kg dry mass and 288 mL volume.	Inferior thermal conductance to 6310G3; retained as a documented comparison, not selected.
Boyd 6310G3	Manufacturer thermal curves at 1.8, 3.7, 7.6 and 15.1 L/min, exact air/liquid pressure-drop curves; 3.9 kg dry mass and 483 mL volume.	Digitized and coupled to the fan, pump and heat-source sensitivity model; fails E3 and is rejected as the final passive-system purchase.
Mishimoto MMDB-KX450F-10LX	Exact SKU and basic construction only.	No thermal or pressure-drop maps; packaging candidate only.

For the selected two-core numerical topology, each 6310G3 receives approximately 5.02 L/min and the nominal fan solution supplies 6.56 m³/min per core. After subtracting the recorded 5 W/K digitization uncertainty from each core, the combined conductance is insufficient to keep the KW26 inlet below 40°C for any declared E3 heat-partition case. The lowest 1.874 kW case predicts 51.9°C at the KW26 inlet. This is a physical and numerical rejection, not a request for more optimistic curve fitting.

No current radiator is approved for final procurement. Package drawings, mounts and hose fittings also remain unresolved, but they do not change the thermal no-go. A replacement concept must first change the E3 heat-rejection boundary—for example by declaring a lower ambient/duty

envelope or adding active refrigeration—then pass the same map-bounded calculation before purchase.

6 Fan and Air-Path Selection: Numerical No-Go

The exact SPAL VA99-ABL315P/R/A/N-101A/SH is now the numerical candidate. SPAL's official brushless catalogue specifies a 280 mm, 12 V module with 950–3,800 rpm control, 2.5 kg mass, IP68/IP6K9K protection and a maximum-speed pressure-flow/current curve tested at 13.0 V, $20\pm5^{\circ}\text{C}$ and 1.18 kg/m^3 air density on an ANSI/AMCA 210-compliant bench.

The selected sensitivity uses one fan per Boyd 6310G3 core. The 0.35 speed ratio is inside the published 950–3,800 rpm control range. Applying the fan affinity laws only over the transformed published curve and multiplying the bare-core pressure drop by 1.5 gives a nominal intersection of $6.56\text{ m}^3/\text{min}$ per core at 45.6 Pa. This is a theoretical reduced-speed curve, not a manufacturer-tested PWM curve.

The result is not robust: reducing fan pressure by the recorded 10 Pa graph reading uncertainty puts the intersection below the transformed published fan-curve range. Higher speed settings move the operating point beyond the Boyd thermal-map limit of $8\text{ m}^3/\text{min}$, where thermal conductance cannot be claimed by extrapolation. The exact fan therefore remains electrically and environmentally characterized but is not approved as an installed air-path solution.

Final fan approval still requires an overlapping measured/PWM P–Q curve, sealed shroud and duct resistance, available low-voltage power and an installed face-flow result. Because the coupled E3 thermal calculation already fails, none of these later measurements authorizes purchase of the current Boyd/SPAL combination as the final vehicle system.

7 Pump and Hydraulic Selection: Marginal Numerical Result

The AMK inverter establishes a minimum 10 L/min loop requirement at 1.5 bar in its rated cooling condition. Each DD5 branch requires at least 4 L/min. Equal motor branches would receive half the total flow, but the numerical screen also applies a 2:1 branch-resistance imbalance.

The Pierburg CWA150 remains a sensitivity candidate. Its identity is manufacturer-confirmed, while the complete 13.5 V, 80°C 50/50 water-glycol curve used here is hosted by Tecomotive rather than Pierburg/Motorservice. The model therefore subtracts 0.1 bar from that pump curve. It combines two parallel Boyd 6310G3 liquid curves, adds 0.03 bar reading uncertainty to the radiator

branch, and sweeps the unknown remainder of the vehicle system as a quadratic loss referenced at 10 L/min.

At the selected conservative 1.5 bar non-radiator loss, the intersection is 10.03 L/min total at 1.61 bar. With a 2:1 motor-branch resistance ratio, the lower-flow DD5 branch receives 4.16 L/min. Both minimums pass, but the margins are only 0.03 L/min total and 0.16 L/min in the weaker motor branch. Raising the non-radiator reference loss to 1.75 bar reduces the mapped total flow below 10 L/min.

The CWA150 is therefore not hydraulically disproved, but it is not robustly approved. The circuit remains sensitive to the unmeasured KW26, DD5 jacket, hose and fitting losses, coolant state and supply voltage. This marginal hydraulic result cannot override the independent E3 thermal no-go, so the pump is not authorized for purchase as part of the current final system.

8 Overflow Containment and Catch Can

FSG 2026 T7.2.2–T7.2.9 is the current compliance baseline. The final catch-can volume, mounting, materials, discharge routing and system sealing must be checked against the entered-event rule revision and the measured filled-system volume. No commercial catch can is selected in this revision.

The required engineering calculation is based on the actual cold fill volume, maximum coolant temperature, coolant expansion, trapped volume and pressure-cap/overflow arrangement. The resulting vessel and hose routing are then inspected on the built car. The former fixed total-loop volume and product safety factors are withdrawn because the radiator, hose lengths and final topology are not yet fixed.

9 Degassing and Header-Tank Strategy

A header tank or swirl pot may improve filling, deaeration and pump-inlet robustness, but it is not demonstrated as an absolute requirement by the present evidence. Its need and geometry depend on the final loop high points, radiator arrangement, pump NPSH requirement and observed air release during filling and operation.

The selection test is practical: fill and bleed the assembled loop, repeat heat-up/cool-down cycles, observe pump cavitation/flow stability and inspect retained air at all high points. A selected vessel must also satisfy final volume, mounting, material and FSG overflow requirements.

Product-specific centrifugal-separation and capacity claims are deferred until their geometry and test evidence are available.

10 Coolant Chemistry, Hoses and Fittings

The legacy statement that FSAE universally permits only deionized water and prohibits additives is removed. FSG 2026 permits plain water, air or oil in T7.2.2, while the AMK manual specifies circulating-water quality guidance (pH 7–9, conductivity limit) and calls for a corrosion inhibitor; it also warns against electrolysis when more noble materials, such as copper, are mixed with its aluminum alloys. The entered-event rules and all component manufacturers must be reconciled before coolant is selected.

No AN size, hose, fitting or adapter is selected in this revision. The final specification must demonstrate: pressure and temperature rating; compatibility with the final coolant and aluminum components; bend radius and routing; G1/4 interface integrity; vibration retention; electrical isolation where required; and measured contribution to the hydraulic system curve. Material choice follows the verified component interfaces and corrosion-control plan rather than generic motorsport preference.

11 Bench Validation Plan

The current Boyd/SPAL/CWA candidate fails the E3 numerical gate and shall not be procured as the final system. The following tests apply only after a replacement concept passes the same coupled screen and receives a conditional procurement decision. They calibrate the numerical design before vehicle release; they are not a mechanism for overriding a pre-procurement numerical no-go.

1. **Hydraulic characterization:** measure total flow, each motor-branch flow and differential pressure versus pump voltage using the final circuit hardware. Compare the measured system curve to the pump curve at the final coolant temperature and chemistry.
2. **Radiator and fan calorimetry:** apply a traceable heater load; measure coolant flow, coolant inlet/outlet temperatures, air inlet/outlet temperatures, air-side differential pressure and fan electrical power. Repeat over the operating-flow range.
3. **Thermal transient:** apply a representative, recorded heat-load sequence and measure component-simulator, coolant and air temperatures. Use the result to identify thermal capacitance and peak-temperature margin.

4. **Fill, bleed and overflow:** perform repeated fill/bleed and heat/cool cycles. Record trapped-air observations, pump-flow stability, overflow volume and leak inspection.

The first acceptance criterion is evidence completeness: every sensor has a calibration/accuracy record and every test has a configuration, ambient condition and raw-data archive. The thermal energy-balance residual target is within 5% after instrumentation uncertainty is quantified. Component limits (flow, temperature, pressure and coolant rise) must be met under the conditions to which their manufacturer specifications apply.

Two targeted component tests close the present powertrain-data gaps. Inverter calorimetry shall compare DC input power, three-phase output power and cold-plate coolant heat under the documented cooling condition. Motor testing shall compare electrical input, shaft power, AMK map total loss and jacket coolant heat at defined speed/torque points. The required equations, instrumentation records and acceptance gates are defined in the internal component-loss characterization procedure.